

Javier Díez Pérez

AI PRODUCT INNOVATION LEAD & AGENTIC SYSTEMS ENGINEER

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CORE SKILLS

AI & Agents
Multi-agent orchestration (Agno, ADK, LangGraph), MCP servers, RAG, Multimodal RAG, GraphRAG, Agentic RAG, RIG, Tool-use architectures, Agent evaluation & observability, Specs-driven development, Human-in-the-loop design, LLM application security, Claude Code plugins

Data Science & ML
scikit-learn, XGBoost, PySpark, Snowpark, Dask, Ray, H2O, SHAP, lime, Pandas, Polars

Knowledge Graphs
RDF/LPG, SPARQL, Neo4j, Ontologies, Causal KGs, Biomedical KGs, GraphRAG, Data Modelling

Programming
Python, R, SQL, Scala, Rust, HTML/CSS

Data Stores
Snowflake, Delta Lake, Unity Catalog, PostgreSQL, MySQL, DuckDB, MongoDB, Hive, Elasticsearch, Vector DBs

Pipelines & CI/CD
dbt, DABs, Prefect, Airflow, Nextflow, Snakemake, GitHub Actions, pytest, Docker

Cloud & DevOps
Databricks Data Engineering (Delta Lake, Unity Catalog, DABs, Agentic RAG), AWS (Big Data & ML), GCP, Terraform, Packer

Bioinformatics
Bioconductor, Biopython, NGS pipelines, GATK, SAMTools, scanpy

Statistics & GIS
Experiment design, multivariate analysis, PostGIS, QGIS

LANGUAGES

PROFILE

Driven by the intersection of science and technology, I operate as a bold innovator who turns complex research into disruptive, scalable realities. I define the product vision, strategy, and roadmap for next-generation, multi-agentic platforms that challenge traditional industry workflows. Operating at the convergence of advanced machine learning, empirical research, and business development, I translate complex multimodal AI capabilities into high-value commercial tools. I am adept at navigating ambiguity, validating user needs in uncharted markets, and managing ethical guardrails around sensitive data. By driving orchestration-focused product design, I ensure we deliver compliant, high-impact products that solve genuine bottlenecks.

EXPERIENCE

Independent consultant - Barcelona
2026 - Present

AI engineer building agentic innovation systems across Biotech amd Healthcare. As an AI-native innovator, I architect innovative, AI-native products from the ground up using Python, Agno, and Google ADK.
While my technology is deeply technical, my philosophy is profoundly people-centric and patient centric. By anchoring LLMs to Knowledge Graphs and Causal Graphs, I ensure that my autonomous agents deliver deterministic, safe, and transparent outcomes that directly accelerate R&D pipelines, optimize clinical trials, and ultimately improve patient care.

AstraZeneca · Barcelona
2023 — 2026

Senior Innovation Specialist DS & AI (2024–2026)

Spearheaded technical vision and assumed cross-functional leadership in designing and delivering agentic AI products, making complex enterprise data actionable for scientific and commercial teams.

- Architected and deployed a production agentic conversational application combining RAG and Retrieval-Interleaved Generation (RIG) using a multi-agent design built on Agno and Google ADK, serving several internal teams.
- Delivered a competitive-intelligence multi-agent application based on the Strategyzer methodology to support strategic business-unit decisions, reducing standardized analysis for key use-cases from weeks or months of manual work to just hours.
- Developed enterprise MCP (Model Context Protocol) servers to expose complex data sources to LLM agents, standardizing agent-to-data interactions across hundreds of data sources.
- Implemented multi-modal RAG pipelines for enterprise knowledge retrieval across text and image assets, reducing search time from days and weeks to minutes.

Data Product Lead — Oncology Data Science Platforms (2023–2025)

Provided strategic and operational leadership for oncology data platforms, driving cross-team collaboration to power translational analytics.

- Led data product operations for patient-derived xenograft (PDx) mouse models and the ENABL platform, ensuring reliable and FAIR data delivery to tens of teams / hundred of individuals analysts, managing several Tb of multiomics data.
- Designed and maintained the oncology data science platform infrastructure to support discovery and translational analytics across tens of programs.

Data Engineer & Python Developer
2022 — 2023

Contractor · Madrid

- Data products for pharma (GSK) and food (HdR); Python packages for data validation & FAIRification
- GCP ETL pipeline development with integrated validation framework

Lead Bioinformatician & Data Scientist
2021 — 2022

PharmaMar · Madrid

- Led bioinformatics R&D strategy and directed cross-functional support for cancer genomics, drug development, and biomarker discovery pipelines across all R&D molecular biology programs.
- Integrated multiomics data (DNA-seq, RNA-seq, ChIP-seq, ATAC-seq) with clinical trial data to enable advanced translational analyses for several patient cohorts for tens of research projects and clinical trials.

Senior Bioinformatician & Data Scientist
2020 — 2021

Fujitsu · Madrid

- Supported the digital transition by mapping clinical data standards to intelligence platforms for the Spanish national drug regulator.
- Built EHR APIs and developed SNOMED-CT mapping pipelines using named-entity recognition to improve healthcare data interoperability, handling thousands of patient HR from most regional healthcare system PRs.

Spanish	Native
English	Proficient
French	Good
Portuguese	Basic

VOLUNTEERING

CRAM Marine Animals Rehab (2024–25)
Madrid4Refugees coding teacher
Save the Children tutor

Senior Bioinformatics Scientist
2019 — 2020
HiFiBio · Paris

- Single cell RNA-seq analysis for TCR in TILs; sc-targeted RNAseq pipeline development
- Algorithm development for cell barcode UMI assignment and sequence correction
- Unsupervised ML data analysis for cell-type and TCR chain-type characterization

Data Analyst & Python Developer
2017 — 2020
Freelancer · Madrid

- Predictive analytics tools for banking, insurance, and telco industries
- Big data science and ML with Python and R ecosystem; package and tool development

Senior Bioinformatician
2016
Wellcome Trust Sanger Institute · Cambridge, UK

- Somatic mutation signatures on DNA repair defective cell lines
- Whole genome sequencing analysis of paired normal/tumor samples: SNV, Indel, CNVs
- Whole exome sequencing analysis from TCGA (The Cancer Genome Atlas)

Bioinformatics Analyst
2015
Congenica Ltd. · Cambridge, UK

- Splicing mutation effect analysis, mutation-created splice sites and cryptic splice sites
- Bioinformatic analysis of trio and family from NGS data

Bioinformatician — Molecular Pathology
2014 — 2015
Royal Marsden Hospital NHS Foundation Trust · London, UK

- NGS pipelines for tumor molecular pathology: panel targeted, immunoglobulin/TCR, amplicon analysis
- Amplicon molecular barcoding deduplication pipeline
- Study Bioinformatician at FoRMAT (molecular characterisation of advanced GI tumors)

Bioinformatician — Clinical Neurogenetics
2013 — 2014
UCL / National Hospital for Neurology · London, UK

- NGS pipelines for clinical neurogenetics: exome, panel targeted, mtDNA heteroplasmy analysis
- Database development for clinical information, diagnostic test results, and annotation

Field Bioinformatics Specialist
2012 — 2013
Life Technologies · Mexico DF, Mexico

- Set up bioinformatic support and training for Ion Torrent (PGM/Proton) and SOLiD platforms across Latin America
- Project design, troubleshooting for whole genome, resequencing, and targeted sequencing

Bioinformatician — Epigenetics & Cancer Biology
2011 — 2012
IDIBELL (Bellvitge Biomedical Research Institute) · Barcelona

- EPINORC project: Epigenetic Disruption of Non-Coding RNAs in Human Cancer (ERC Advanced Grant)
- Methylation arrays, bisulphite sequencing, WGS, exome NGS, RNA-seq, ChIP-seq analysis

Bioinformatics Research Assistant
2008 — 2010
CRG / Parc de Recerca Biomèdica de Barcelona

- De novo genome assembly, resequencing and genome analysis of Saccharomyces cerevisiae
- Developed DeathBase: relational database and web front-end for cell death curated proteins
- Characterization of putative human BH3-only proteins; cancer susceptibility gene analysis

Research Assistant
2005 — 2008
CSIC (CNB & CBM-SO) · Madrid

- Human centrosome proteome study and CentrosomeDB database development
- Cell death genetics in Drosophila; ML approach for phenotype trait analysis

EDUCATION

Specialist — Quantitative Research & Statistics
2006 — 2007
Polytechnic University of Madrid & CSIC

Diploma de Estudios Avanzados (DEA) — Biochemistry & Molecular Biology
2005 — 2007
Universidad Autónoma de Madrid

- **English Designation:** Postgraduate Advanced Research Qualification / Ph.D. Candidate (ABD Status)
- **Framework Tier:** Pre-Bologna qualification classified under ISCED Level 8 / EQF Level 8 (Doctoral Level Framework)
- **Scope:** Completed 60+ ECTS doctoral coursework & defended thesis (*Tesina*) to achieve official Research Proficiency (*Suficiencia Investigadora*)

Licenciatura in Biochemistry (M.Sc. Equivalent)
2001 — 2004
Universidad Autónoma de Madrid
Equivalence: MECES Level 3 (Máster) & EQF Level 7 · National Registry No. 2005015238

SELECTED PUBLICATIONS

PEER-REVIEWED PUBLICATIONS

- Costanzo F, Martínez Diez M, Santamaría Nuñez G, Díaz-Hernández JI, Genes Robles CM, **Díez Pérez J**, Compe E, Ricci R, Li TK, Coin F, Martínez Leal JF, Garrido-Martín EM, Egly JM. "Promoters of ASCL1- and NEUROD1-dependent genes are specific targets of lurbinectedin in SCLC cells." *EMBO Mol Med*, 2022. PMID:35263037
- Jiménez-Guardeño JM, Ortega-Prieto AM, Menéndez Moreno B, Maguire TJA, Richardson A, Díaz-Hernández JI, **Diez Perez J**, Zuckerman M, et al. "Drug repurposing based on a quantum-inspired method versus classical fingerprinting uncovers potential antivirals against SARS-CoV-2." *PLoS Comput Biol*, 2022. PMID:35849631
- Glodzik D, Morganella S, Davies H, ..., **Diez-Perez J**, ..., Nik-Zainal S. "A somatic-mutational process recurrently duplicates germline susceptibility loci and tissue-specific super-enhancers in breast cancers." *Nature Genetics*, 2017. PMID:28112740
- Galindo-Moreno J, Iurlaro R, El Mjiyad N, **Díez-Pérez J**, Gabaldón T, Muñoz-Pinedo C. "Apolipoprotein L2 contains a BH3-like domain but it does not behave as a BH3-only protein." *Cell Death Dis*, 2014. PMID:24901046
- Heyn H, Li N, Ferreira HJ, Moran S, Pisano DG, Gomez A, **Diez J**, ..., Esteller M. "Distinct DNA methylomes of newborns and centenarians." *PNAS*, 2012. PMID:22689993
- Neurohr G, Naegeli A, Titos I, Theler D, Greber B, **Díez J**, Gabaldón T, Mendoza M, Barral Y. "A midzone-based ruler adjusts chromosome compaction to anaphase spindle length." *Science*, 2011. PMID:21393511
- **Díez J**, Walter D, Muñoz-Pinedo C, Gabaldón T. "DeathBase: a database on structure, evolution and function of proteins involved in apoptosis and other forms of cell death." *Cell Death Differ*, 2010. PMID:20383157
- Nogales-Cadenas R, Abascal F, **Díez-Pérez J**, Carazo JM, Pascual-Montano A. "CentrosomeDB: a human centrosomal proteins database." *Nucleic Acids Res*, 2009. PMID:18971254
- Bonifaci N, Berenguer A, **Díez J**, Reina O, Medina I, Dopazo J, Moreno V, Pujana MA. "Biological processes, properties and molecular wiring diagrams of candidate low-penetrance breast cancer susceptibility genes." *BMC Med Genomics*, 2008. PMID:19094230

PRESENTATIONS & CONFERENCES

- Parmar C, Roudko V, Muckerson V, Blando J, Daniel J, **Díez J**, Gunda S, Mendoza Alcala A, Franco A, Stambolic I, Benchea C, Stewart R, Lai Z, Reis-Filho J, Scaltriti M, Palmer D. "AI-Derived H&E Features Stratify Overall Survival and Suggest Differential Benefits in NEPTUNE (1L mNSCLC)." *AACR Annual Meeting*, San Diego, 2026
- Moorcraft SY, González de Castro D, Cunningham D, Walker BA, Hulkki Wilson S, **Díez Pérez J**, et al. "FOrMAT: Feasibility of a molecular characterization approach to treatment of patients with advanced GI tumors." *J Clin Oncol* 33, 2015 (ASCO). DOI
- McCall S, Polke JM, **Díez-Pérez J**, Gardiner A, Jaffer F, Nanji T, Pittman A, Hughes D, Hanna MG, Houlden H. "Next Generation Sequencing of Neuronal Channelopathies." *British Society for Genetic Medicine Annual Meeting*, Liverpool, 2013
- "ION PROTON DATA ANALYSIS" — Mexico & Latin America Ion Proton launch events, Life Technologies, 2012–2013
- ASHG Annual Meeting 2012, San Francisco · RECOMB::DREAM 2011, Bellvitge · Mono-allelic Expression in Health and Disease, Barcelona · Systems Biology & New Sequencing Technologies (SBNST), CRG Barcelona

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